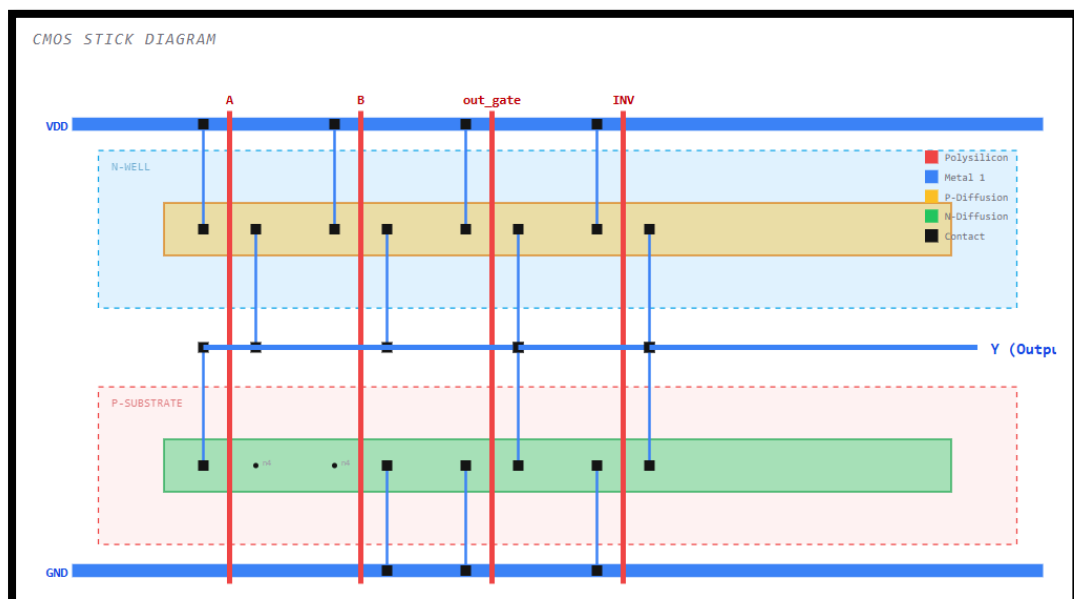
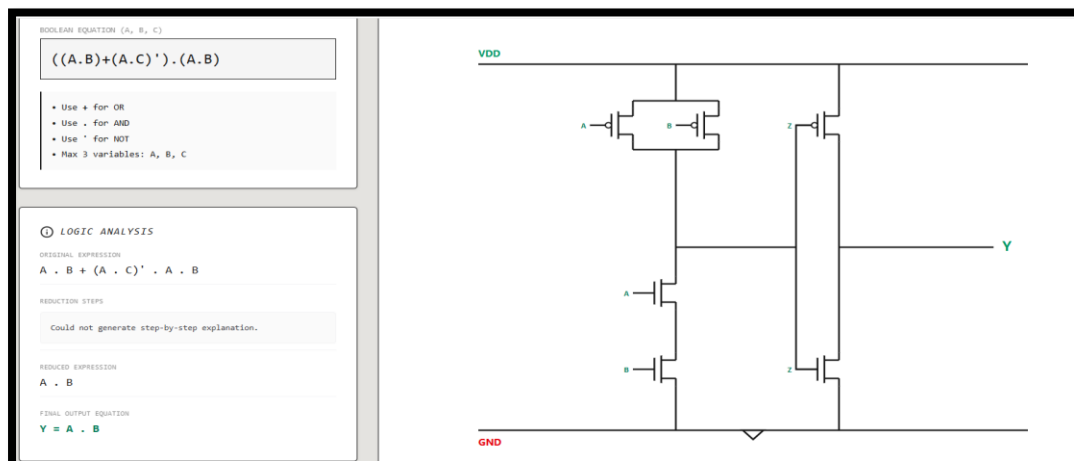


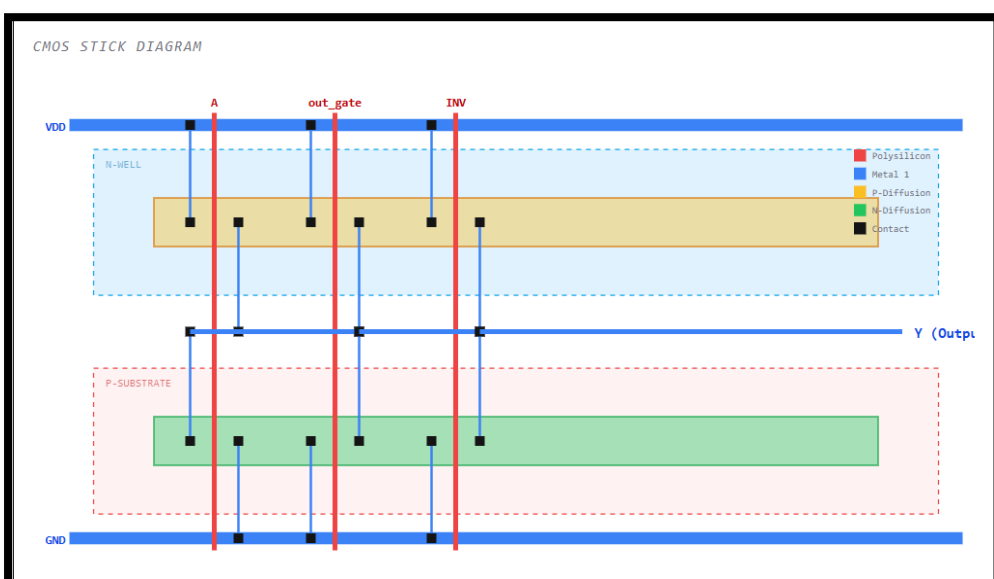
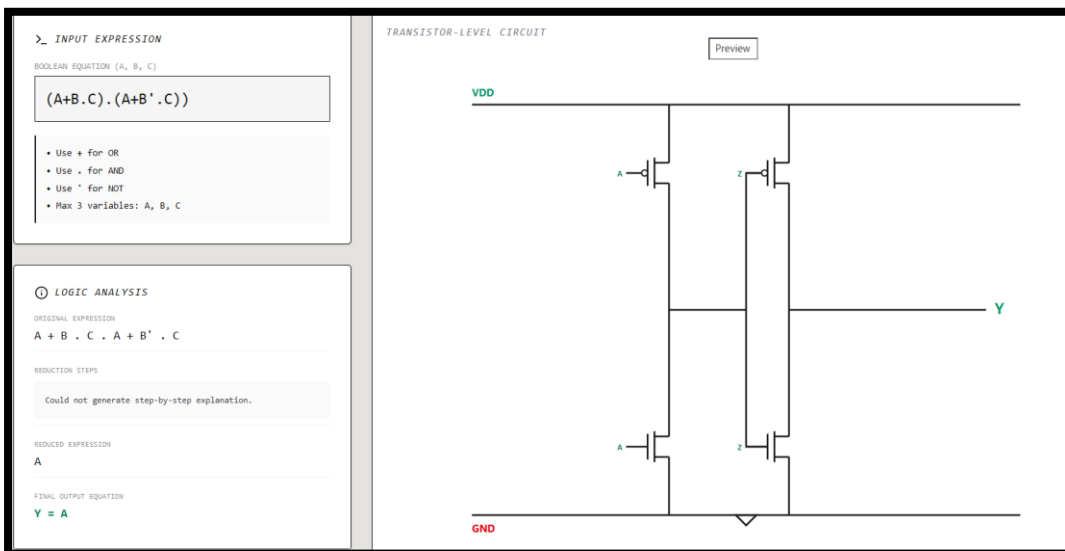
$$1. F = ((A.B) + (A.C)') . (A.B)$$

$$\begin{aligned}
 1) \quad F &= ((A.B) + (\overline{C.A})) . (A.B) \\
 &= (AB + (\bar{A} + \bar{C})) (AB) \\
 &= (AABB) + (\bar{A}AB) + (ABC\bar{C}) \\
 &= AB + AB\bar{C} \\
 &= AB(1 + \bar{C}) \\
 &= AB \\
 F &= A B
 \end{aligned}$$



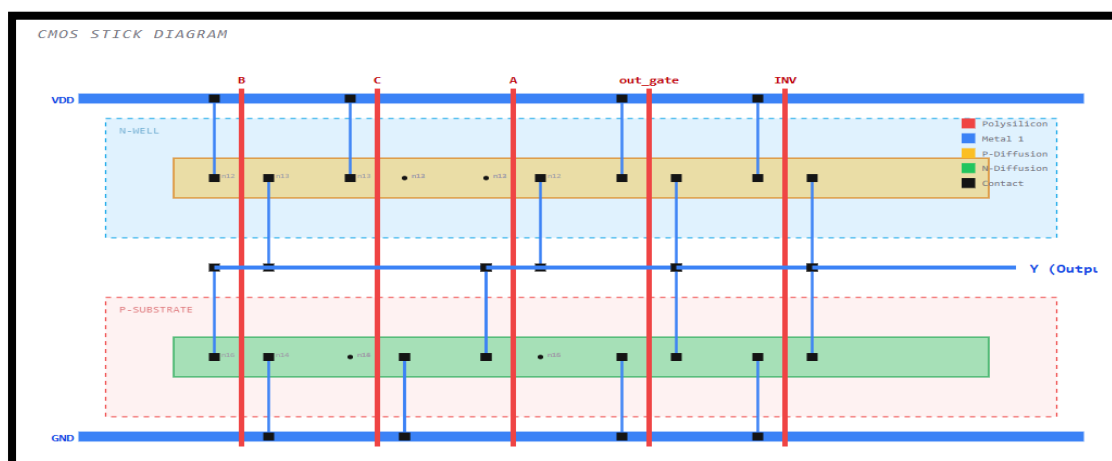
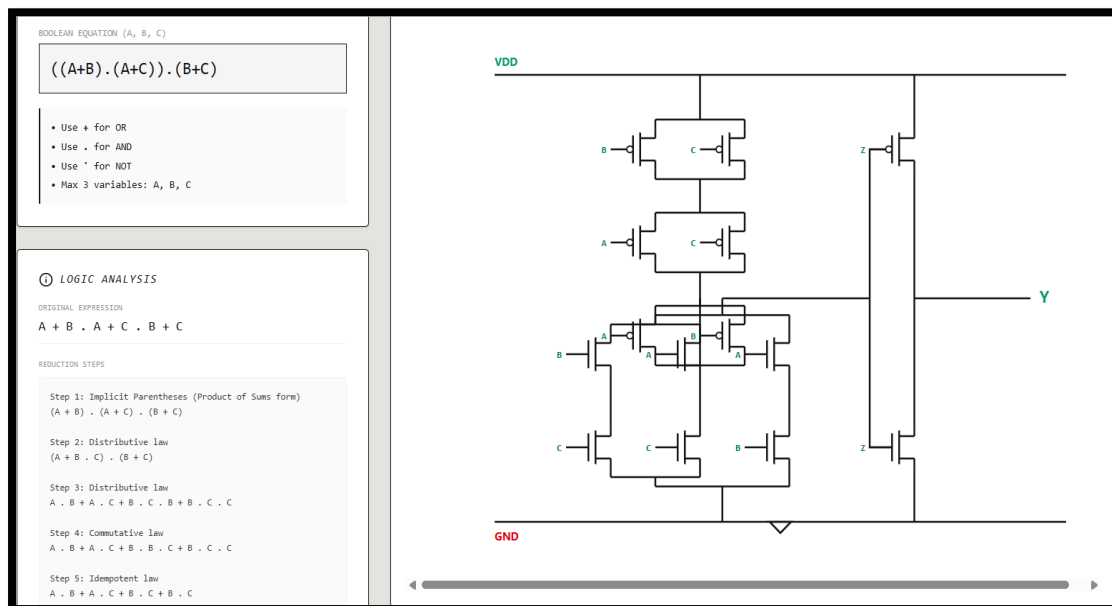
$$2. F = (A+B.C). (A+B'.C)$$

$$\begin{aligned}
 F &= (A+BC)(A+\bar{B}\bar{C}) \\
 &= AA + A\bar{B}\bar{C} + ABC + B\bar{B}\bar{C} \\
 &= AA + A\bar{B}\bar{C} + ABC \\
 &= A + AC(\bar{B} + B) \\
 &= A + AC \\
 &= A(1+C) \\
 &= A
 \end{aligned}$$



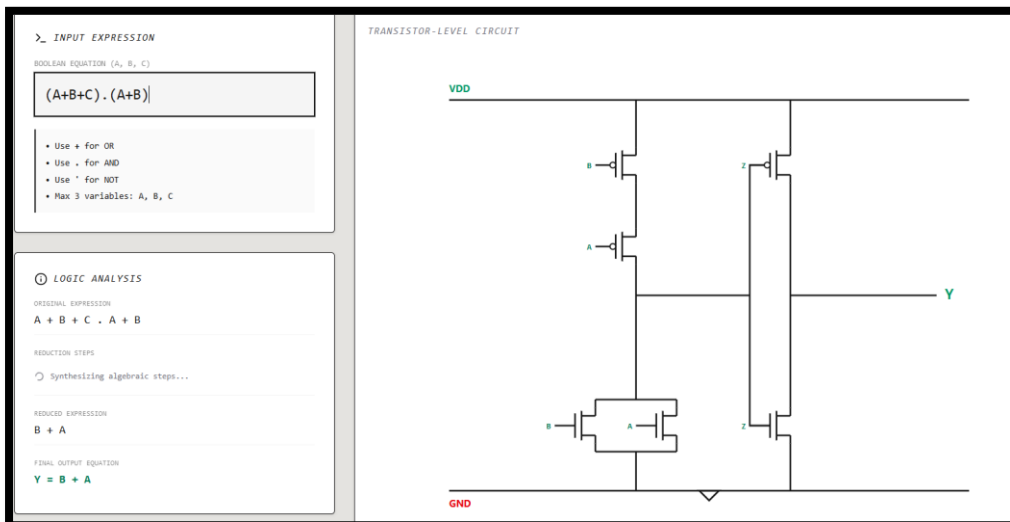
3. $F = ((A+B) \cdot (A+C)) \cdot ((B+C))$

$$\begin{aligned}
 2) F &= ((A+B) \cdot (A+C)) \cdot ((B+C)) \\
 &= (AA + AC + AB + BC) \cdot (B+C) \\
 &= AAB + AAC + ABC + ACC \\
 &\quad + ABB + ABC + BBC + BCC \\
 &= AB + AC + ABC + AC + BC + AB \\
 &= AB + AC + BC + ABC \\
 &= AB(1+C) + C(A+B) \\
 &= ABC + AC + AB \\
 &= AC + BC + AB
 \end{aligned}$$

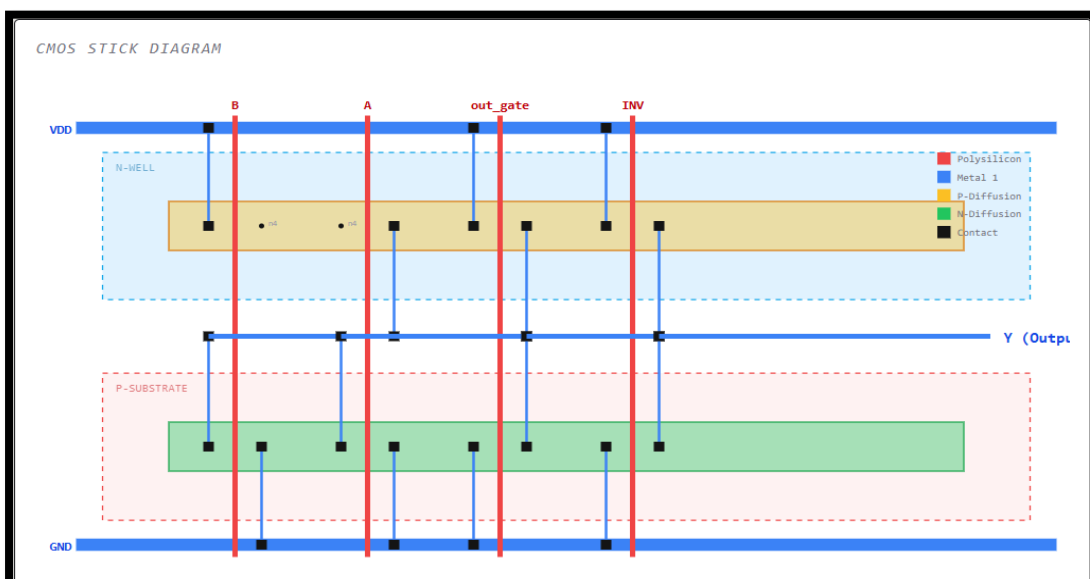


4. $F = (A+B+C) \cdot (A+B)$

$$\begin{aligned}
 F &= (A+B+C) (A+B) \\
 &= AA + AB + BC + AC \\
 &\quad + AB + BB \\
 &= A + AB + B + BC + AC \\
 &= A(1+B) + B(1+C) + AC \\
 &= A + B + AC \\
 &= A(1+C) + B \\
 &= A + B
 \end{aligned}$$

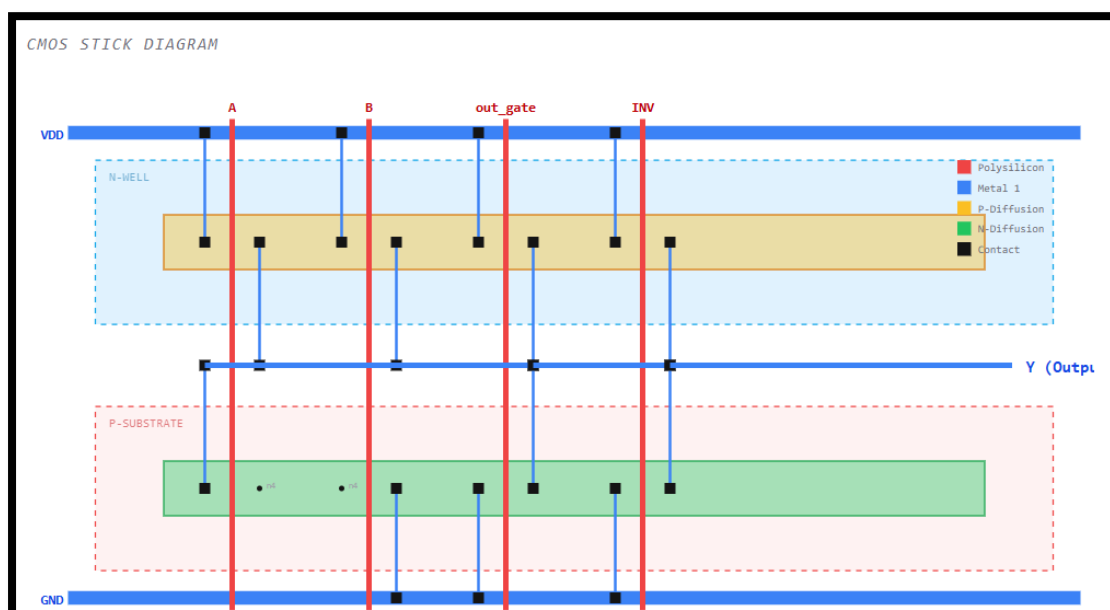
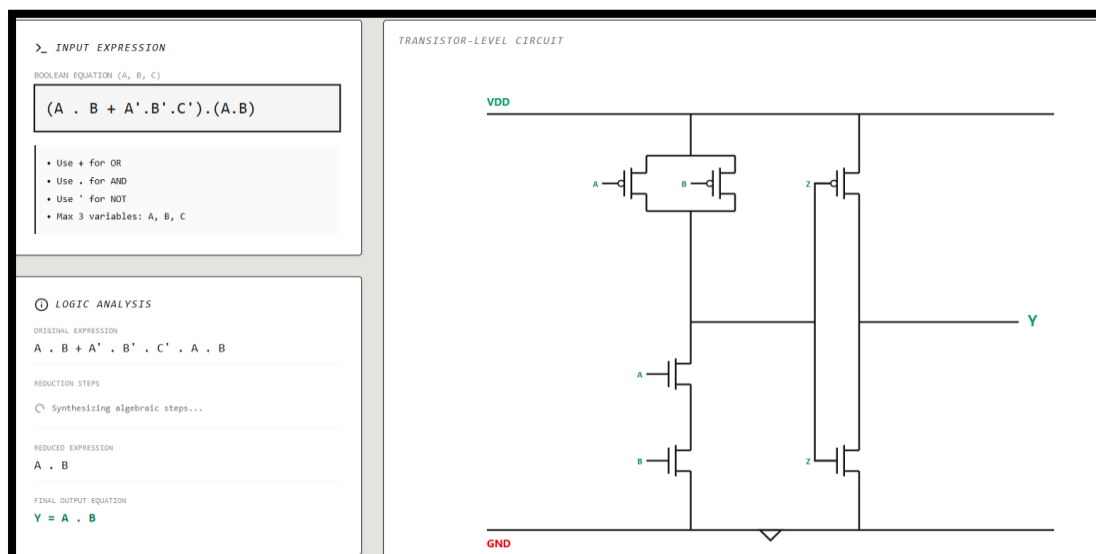


5. $F =$



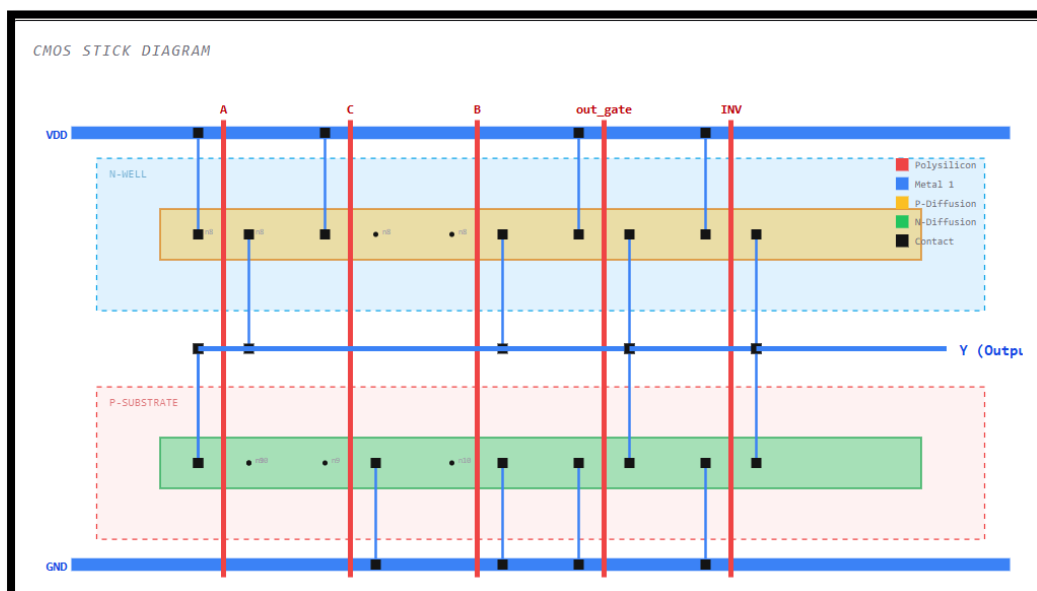
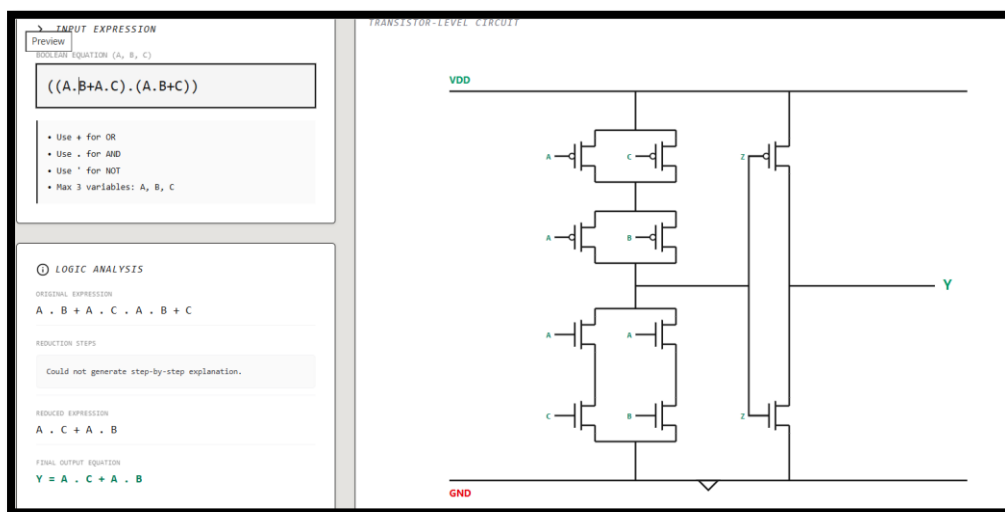
$$(A \cdot B + A' \cdot B' \cdot C') \cdot (A \cdot B)$$

$$\begin{aligned} 3] \quad F &= (AB + \bar{A}\bar{B}\bar{C}) \cdot (AB) \\ &= ABAB + A\bar{A}B\bar{B}\bar{C} \\ &= AB + 0 \\ &= AB \\ F &= AB \end{aligned}$$



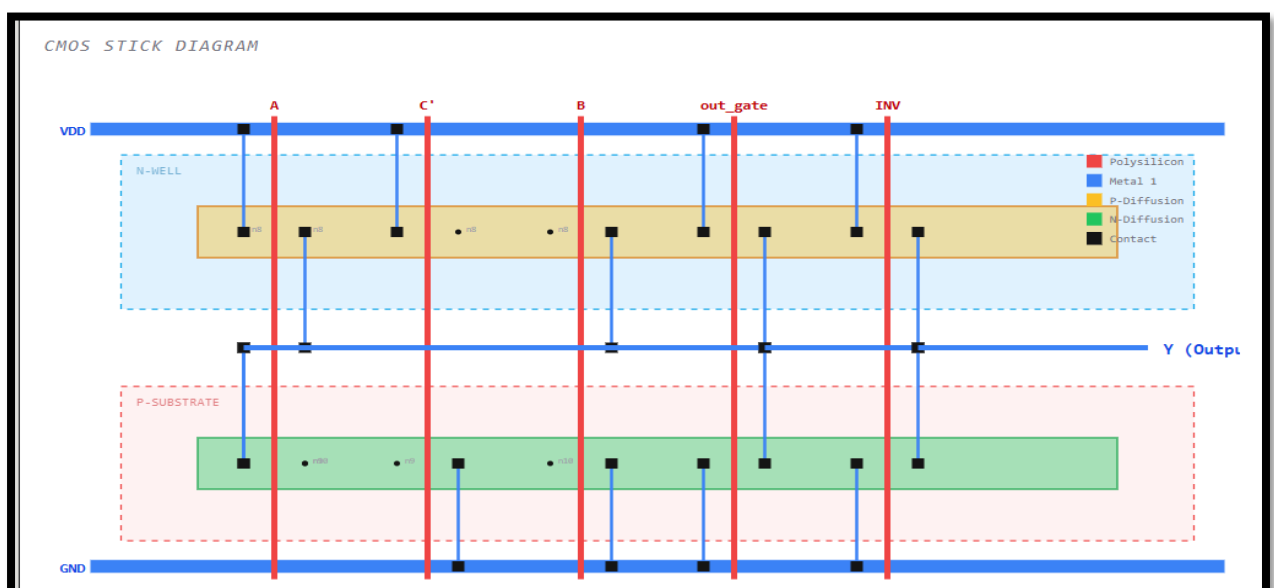
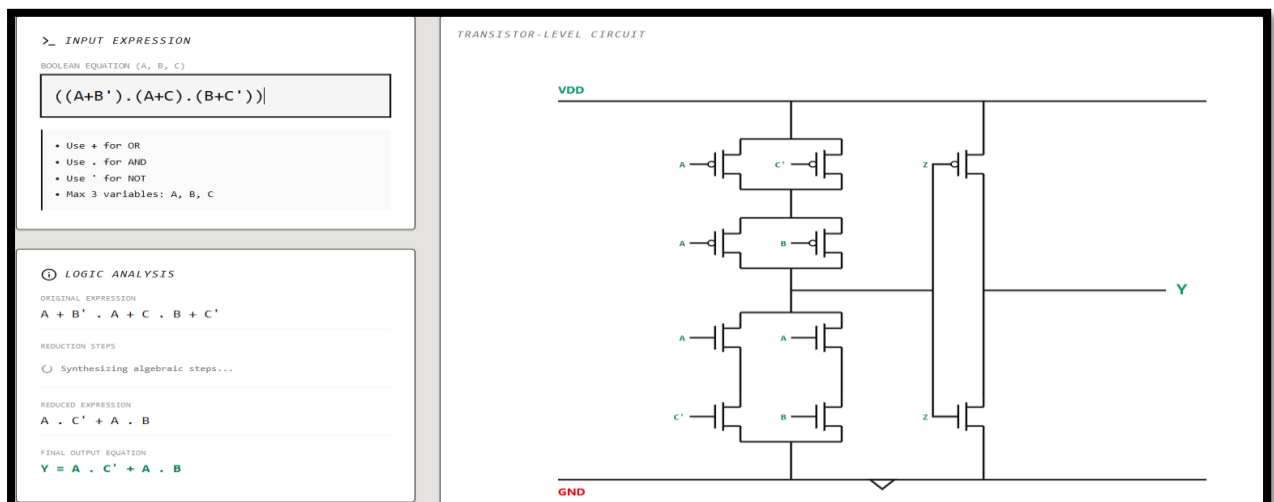
6. $F = ((A \cdot B + A \cdot C) \cdot (A \cdot B + C))$

$$\begin{aligned}
 F &= ((AB + AC) (AB + C)) \\
 &= AAB + ABC + AABC + ACC \\
 &= AB + ABC + ABC + AC \\
 &= AB + AC + ABC \\
 &= AB(1 + C) + AC \\
 &= AB + AC
 \end{aligned}$$



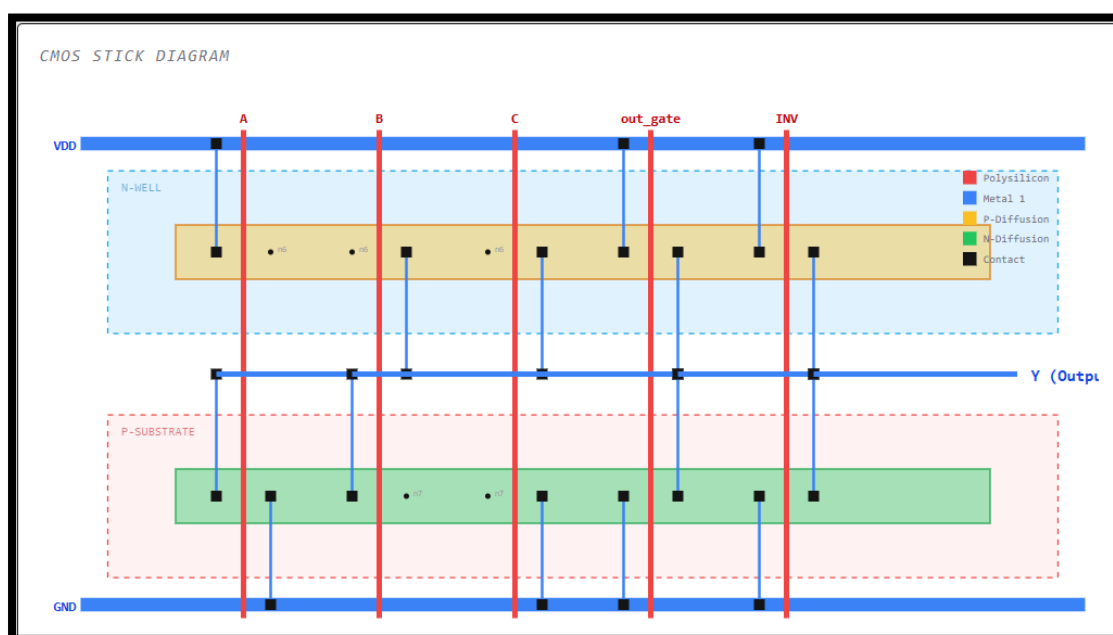
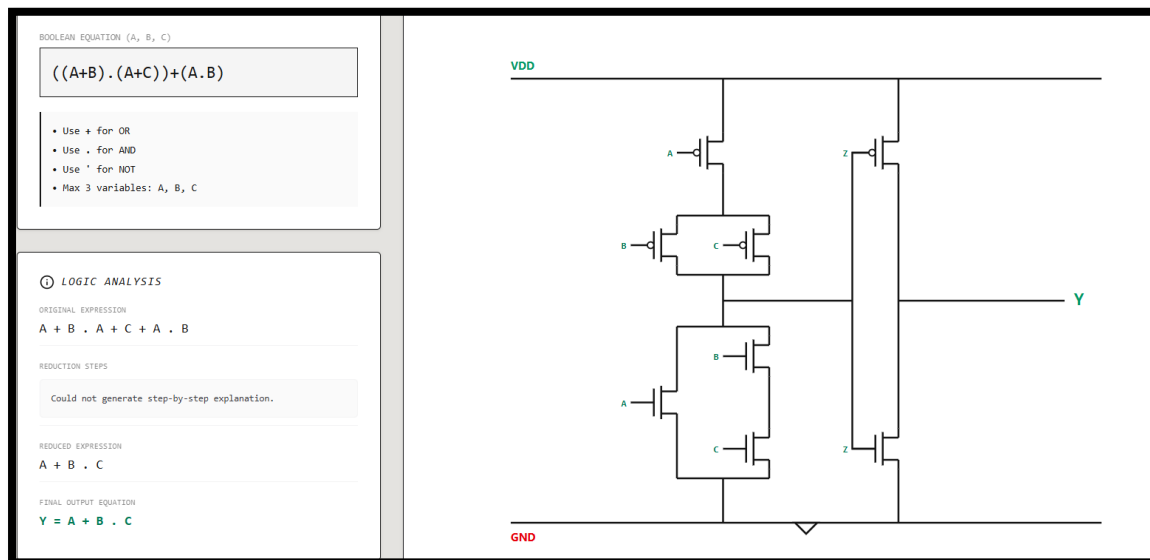
7. $F = ((A+B'). (A+C). (B+C'))$

$$\begin{aligned}
 F &= ((A+B)(A+C)(B+C)) \\
 &= (AA+AC+AB+BC)(B+C) \\
 &= AAB+ACB+A\bar{B}B+B\bar{B}C \\
 &\quad +AA\bar{C}+A\bar{A}C+AB\bar{C} \\
 &\quad +\bar{B}C\bar{C} \\
 &= AB+AC+ABC+AB\bar{C} \\
 &= A\bar{B}+AC+\bar{B}\bar{C} \\
 F &= A\bar{B}+AC+\bar{B}\bar{C} \\
 F &= A\cdot\bar{C}+A\cdot B
 \end{aligned}$$



8. $F = ((A+B) \cdot (A+C)) + (A \cdot B)$

$$\begin{aligned}
 \textcircled{G} \quad F &= ((A+B)(A+C)) + (AB) \\
 &= AA + AC + AB + BC + AB \\
 &= A + AC + AB + BC + AB \\
 &= A(1+C) + AB + BC \\
 &= A + AB + BC \\
 &= A(1+B) + BC \\
 &= A + BC
 \end{aligned}$$



9. $F = A.B + (A+C). (A.B)'$

$$\begin{aligned} F &= AB + (A+C). \overline{AB} \\ &= AB + (A+C). (\overline{A} + \overline{B}) \\ &= AB + A\overline{A} + A\overline{B} + \overline{A}C \\ &\quad + \overline{B}C \\ &= AB + A\overline{B} + \overline{A}C + \overline{B}C \\ &\quad + \cancel{A\overline{A}}^0 \\ &= AB + \overline{A}C + A\overline{B} + \overline{B}C \\ &= A(B + \overline{B}) + \overline{A}C + \overline{B}C \\ &= A + \overline{A}C + \overline{B}C \end{aligned}$$

10. $F = (A+A')(B+B')$

$$\begin{aligned}
 F &= (A+\bar{A})(B+\bar{B}) \\
 &= AB + A\bar{B} + \bar{A}B + \bar{A}\bar{B} \\
 &= A(B+\bar{B}) + \bar{A}(B+\bar{B}) \\
 &= A + \bar{A} \\
 F &= 1
 \end{aligned}$$

> INPUT EXPRESSION

BOOLEAN EQUATION (A, B, C)

$(A+A') \cdot (B+B')$

- Use + for OR
- Use . for AND
- Use ' for NOT
- Max 3 variables: A, B, C

LOGIC ANALYSIS

ORIGINAL EXPRESSION

$A + A' \cdot B + B'$

REDUCTION STEPS

Could not generate step-by-step explanation.

TRANSISTOR-LEVEL CIRCUIT

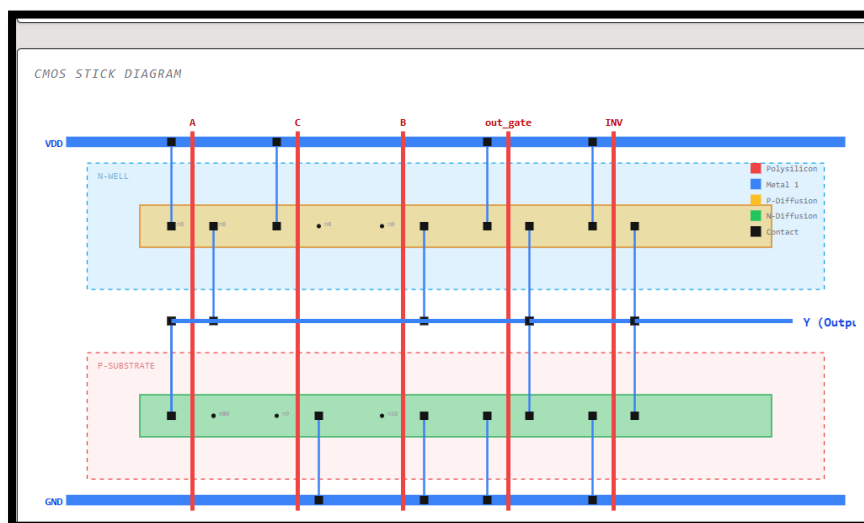
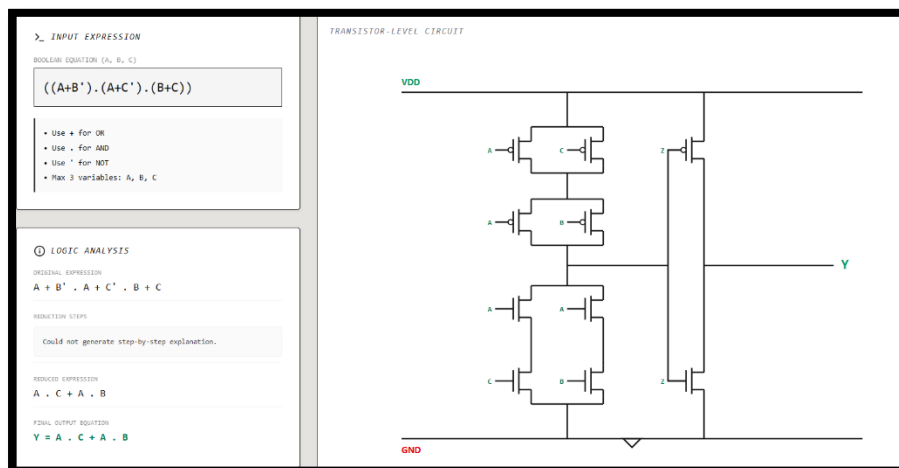
$Y = \overline{VDD}$

CMOS STICK DIAGRAM

No Stick Diagram Required (Constant Logic)

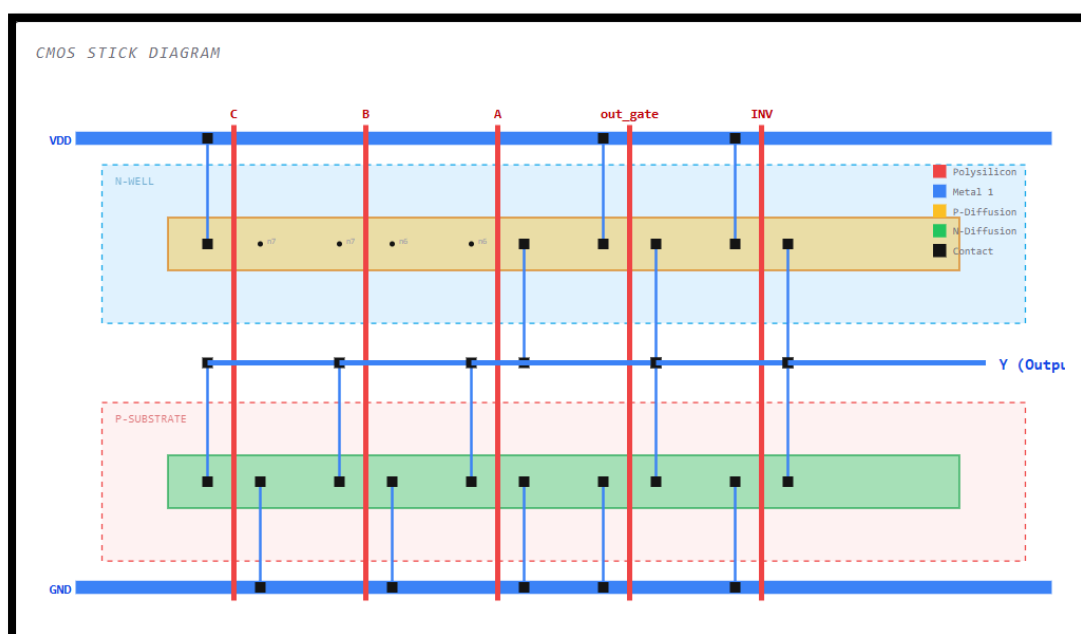
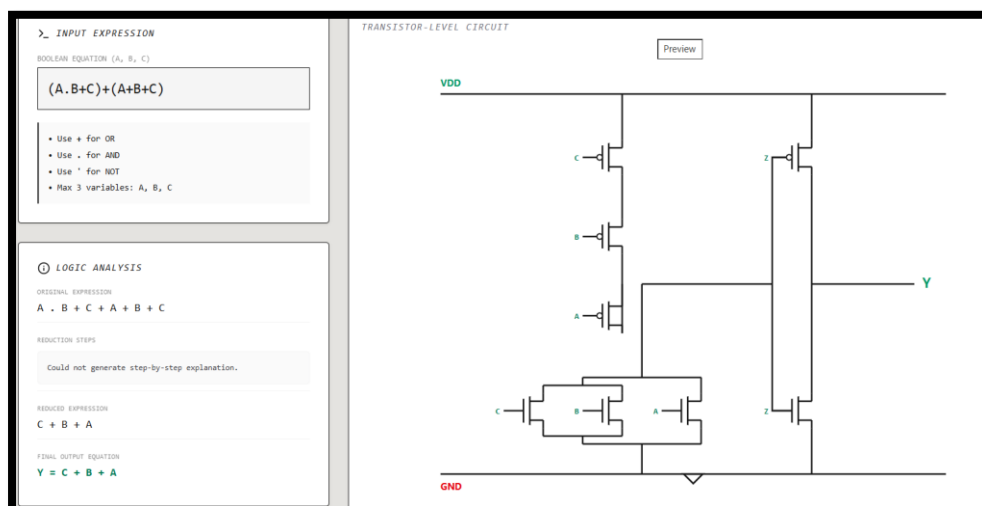
11. $F = ((A+B') \cdot (A+C')) \cdot (B+C)$

$$\begin{aligned}
 F &= ((A+B') \cdot (A+C')) \cdot (B+C) \\
 &= (A \cdot A + A \cdot \bar{C} + A \bar{B} + \bar{B} \bar{C}) \cdot (B+C) \\
 &= AAB + A\bar{C}B + A\bar{B}\bar{C} + \bar{B}\bar{C}C + AAC + A\bar{B}C + A\bar{B}C + \bar{B}\bar{C}C \\
 &= AB + AC + A\bar{B}\bar{C} + A\bar{B}C \\
 &= AB(1 + \bar{C}) + AC(1 + \bar{B}) \\
 &= AB + AC
 \end{aligned}$$



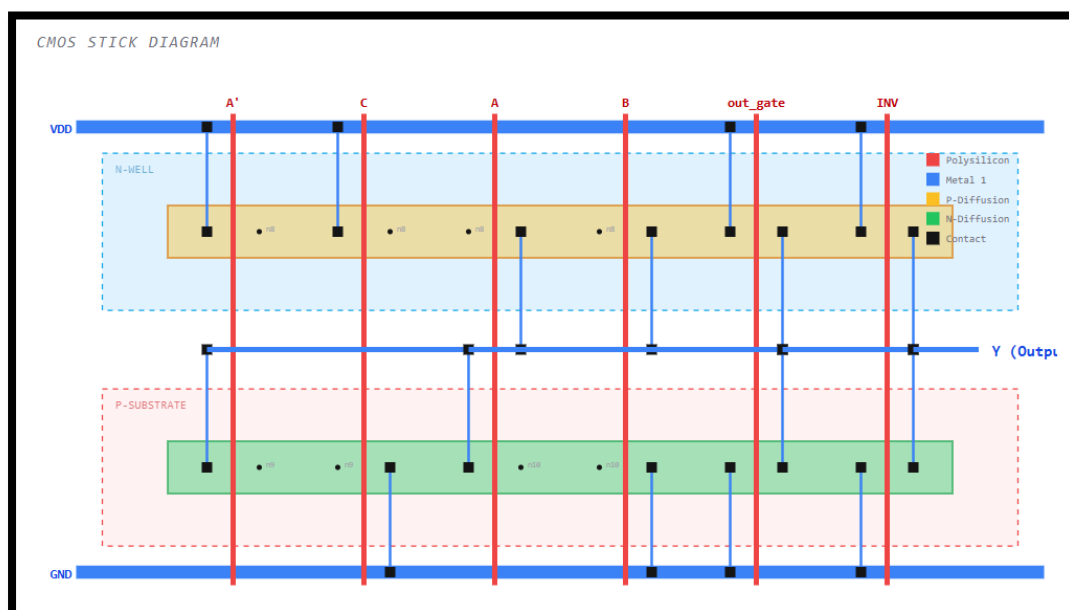
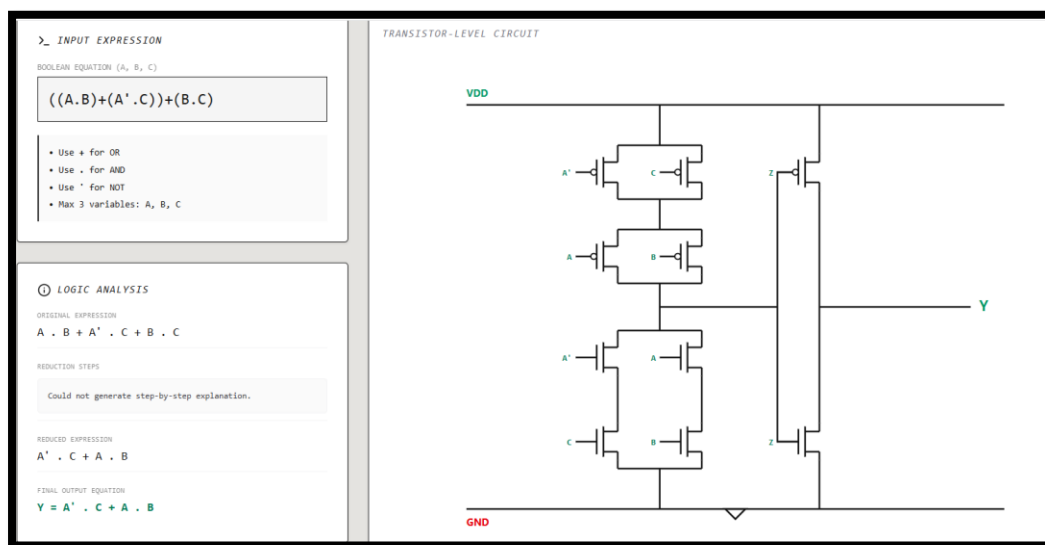
12. $F = (A \cdot B + C) + (A + B + C)$

14) $F = (AB + C) + (A + B + C)$
 $= ABA + ABB + ABC + AC + BC + CC$
 $= AB + AB + AC + BC + ABC + C$
 $= AB(1 + C) + AC + BC + C$
 $= AB + C(1 + A) + BC$
 $= AB + C(1 + B)$
 $F = \underline{AB + C} = \underline{A + B + C}$



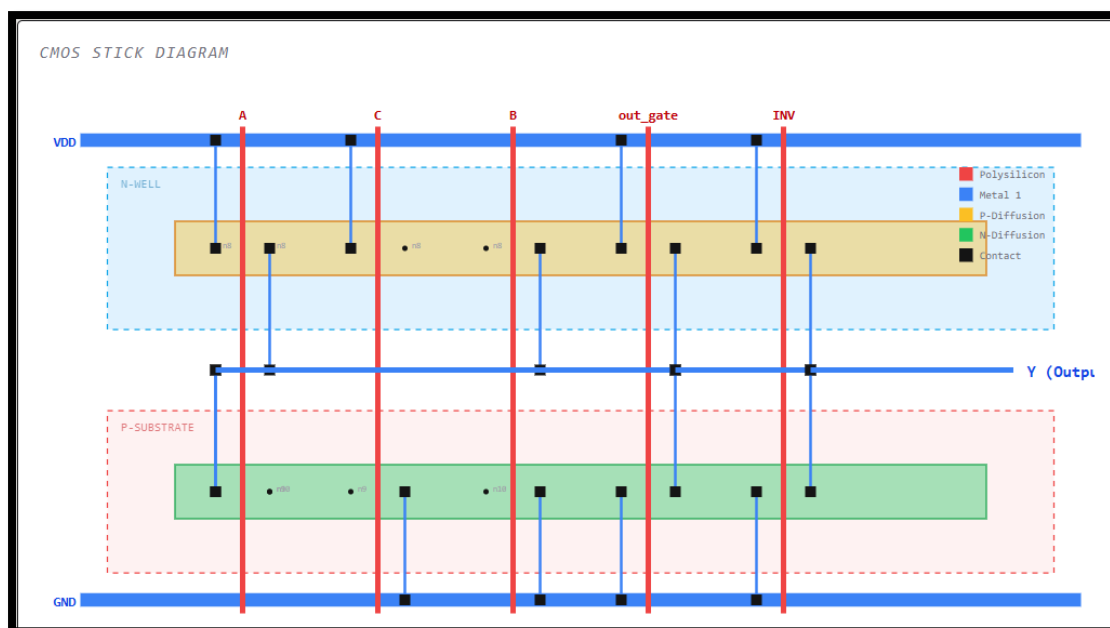
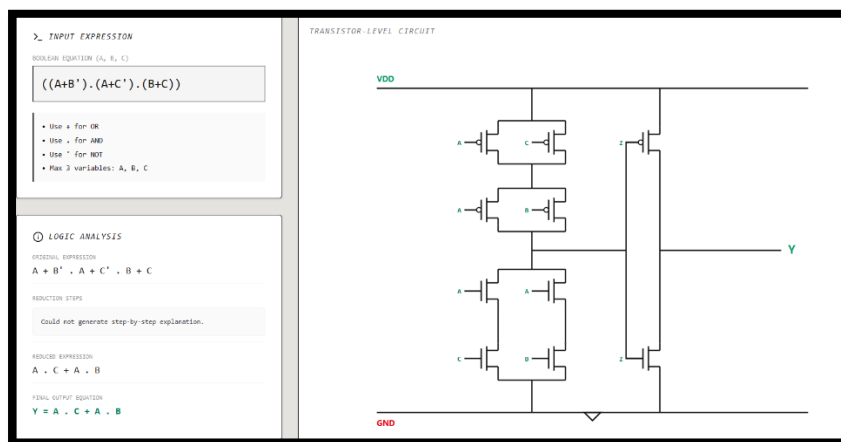
13. $F = ((A.B) + (A'.C) + (B.C))$

$$\begin{aligned}
 F &= ((A.B) + (\bar{A}.C) + (B.C)) \\
 &= AB + \bar{A}C + BC \\
 &= B(A + C) + \bar{A}C \\
 &= \bar{A}CC + AB \\
 &= AB + \bar{A}C
 \end{aligned}$$



14. $F = ((A+B'). (A+C')). (B+C)$

$$\begin{aligned}
 F &= ((A+B)(A+C)(B+C)) \\
 &= (AA + A\bar{C} + A\bar{B} + \bar{B}\bar{C})(B+C) \\
 &= AAB + A\bar{C}B + A\bar{B}B + \bar{B}\bar{C}B \\
 &\quad + AAC + A\bar{C}\bar{C} + A\bar{B}\bar{C} + \bar{B}\bar{C}\bar{C} \\
 &= AB + AC + A\bar{B}\bar{C} + \bar{B}\bar{C} \\
 &= AB(1+\bar{C}) + AC + \bar{B}\bar{C} \\
 &= AB + AC
 \end{aligned}$$



15. $F = (A+B+C) \cdot (A+B+C)'$

$$\begin{aligned}
 \text{ii) } F &= (A+B+C)(\overline{A+B+C}) \\
 &= (A+B+C)(\bar{A}\bar{B}\bar{C}) \\
 &= A\bar{A}\bar{B}\bar{C} + B\bar{A}\bar{B}\bar{C} + C\bar{A}\bar{B}\bar{C} \\
 &= 0 + 0 + 0 \\
 F &= 0
 \end{aligned}$$

> INPUT EXPRESSION

BOOLEAN EQUATION (A, B, C)

(A+B+C) · (A+B+C)'

- Use + for OR
- Use · for AND
- Use ' for NOT
- Max 3 variables: A, B, C

LOGIC ANALYSIS

ORIGINAL EXPRESSION

A + B + C · (A + B + C)'

REDUCTION STEPS

↳ Synthesizing algebraic steps...

REDUCED EXPRESSION

0

FINAL OUTPUT EQUATION

Y = 0

TRANSISTOR-LEVEL CIRCUIT

Y = GND

↓

PDN (NMOS Network):

- Logic: 0
- AND $\rightarrow$ Series, OR $\rightarrow$ Parallel

PUN (PMOS Network):

- Logic: Dual of PDN
- AND $\rightarrow$ Parallel, OR $\rightarrow$ Series

CMOS STICK DIAGRAM

No Stick Diagram Required (Constant Logic)